

The Role of Human Agency in Human-AI Co-Creativity

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Large language models (LLMs) are popular tools for creative ideation, but have been shown to homogenize outputs across users. We test if a high-agency mindset—approaching AI to support, rather than replace, human creative capabilities—can mitigate this homogenization effect. Participants were experimentally assigned to a high-agency mindset with AI access, low-agency mindset with AI access, or a human-only control condition to generate creative uses for everyday objects. Contrary to our expectations, ideas produced by participants in the mindset conditions were rated by third-party evaluators as no more creative than ideas produced by participants in the control condition. However, an exploratory cluster analysis revealed robust differences between the three conditions, whereby the control condition produced the greatest number of unique categories and the low-agency condition showed a single, dominant cluster, while in the high-agency condition ideas were more diffuse across clusters. While AI has enabled greater ideational fluency, our research suggests the degree of agency with which people approach their AI tool has downstream consequences on collective creativity.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI)**; *Empirical studies in HCI*; *Ubiquitous and mobile computing systems and tools*.

Additional Key Words and Phrases: human-AI collaboration, agency, collective creativity, ideation

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1 Introduction

One of the cornerstones of human cognition and communication is our ability to engage in creative ideation: the process of generating, developing and conveying new ideas. People have long relied on tools to support their creative processes, from physical notebooks to mind-mapping software [5]. One of the newest and most popular tools to stimulate creativity are large language models (LLMs), which can provide a thought partner to anyone with an internet connection. Fundamentally, LLMs are token predictors that output likely words given a user prompt and context. Their ability to rapidly synthesize massive amounts of text make them useful for a wide range of creativity tasks, and recent reports suggest that people regularly use these systems for tasks related to creativity [12].

The process of people using LLM systems for ideating and communicating creative ideas we refer to as *human-AI co-creativity*. Research to date suggests two important patterns in human-AI co-creativity. The first is that LLM systems perform as well as or even better than humans on commonly accepted creative ideation benchmarks [7] and benchmarks for ideation in specific contexts [6]. However, when humans use AI systems to be creative, a second pattern emerges: Multiple studies have documented a homogenizing effect of LLMs on people’s creative output, in which people using AI tend to produce more similar outputs compared to their counterparts who do not use AI. This effect has been shown in a range of contexts involving open-ended output, such as product design [1, 15] and short-story

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writing [3]. In recent years computer scientists have termed this paradox *algorithmic monoculture* [8], or, in the case of generative AI, *generative monoculture* [19]. Derived from the notion of monoculture in agriculture, these terms capture the vulnerabilities incurred by mass reliance on a single algorithm. While each AI-assisted individual may generate an idea more creative than what they could achieve on their own, this comes at a risk on the collective level. What is creative with respect to an individual may in fact be banal with respect to a group of individuals using the same AI assistant.

In the present study, we examine whether mindsets—the beliefs that guide people’s interpretations of their experiences in the world—can buffer against the homogenizing effects of LLMs while preserving the benefits of human-AI co-creativity. Specifically we explore whether an *agentic mindset*, in which a person approaches a tool in a way that supports—rather than replaces—their own agency and control. When people exercise an agentic mindset with other technologies, such as social media, they believe they can control the amount and nature of their use [9]. Encouraging individuals to develop a more intentional orientation towards their AI use may lead them to engage more in creative ideation and increase their likelihood of contributing diverse output.

We test the importance of a high-agency mindset in human-AI co-creativity by experimentally manipulating the guidance people receive about AI as a creativity support tool. We compare a group of people who are prohibited from using AI against two treatment groups that were told to use AI: a low-agency condition, in which participants are told to let AI do the bulk of the creative ideation, and a high-agency condition, in which participants are told to leverage their own human creativity in tandem with the AI to develop even better ideas. We assess creativity at both the individual and group levels to explore the effect of agentic mindsets on both the creativity and diversity of ideas produced in human-AI collaboration, and we examine how mindsets influence the behavioral interactions in the human-AI collaboration.

2 Method

2.1 Participants

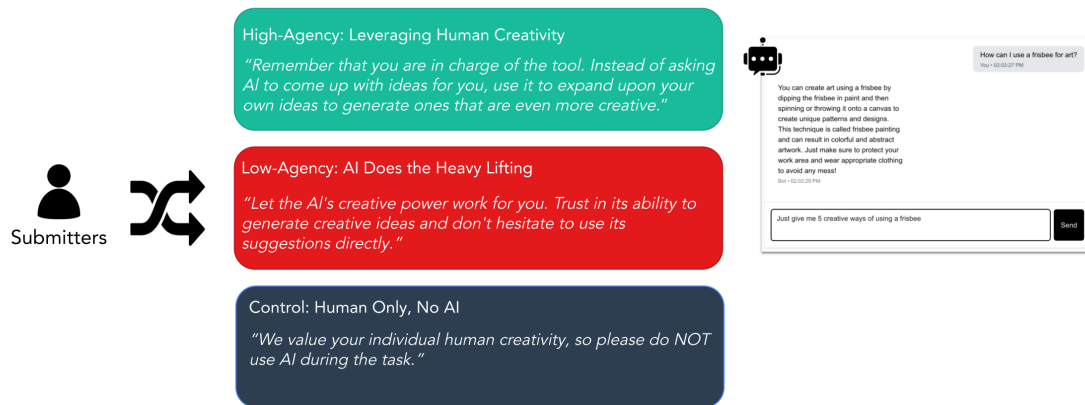
We recruited two samples of adult participants from the U.S. to participate in the study: a sample of submitters and a sample of evaluators. Idea submitters were community college students in the San Francisco Bay Area ($N = 150$; age: $M = 25.1$ years, $SD = 9.2$; 72% female) who participated for research credits. An additional three submitters were excluded for failing an initial attention check. Evaluators were CloudResearch participants ($N = 142$; age: $M = 46.7$ years, $SD = 16.3$; 49% female) who were compensated financially for their time. An additional six participants were excluded for failing an initial attention check and five participants were excluded due to idea data logging issues.

2.2 Procedure

Methods were approved by the [ANONYMIZED] Institutional Review Board. All participants provided informed consent.

2.2.1 Submission Phase. Upon consent, participants were asked to read a general description of the Alternative Uses Task (AUT), in which they would be asked to come up with non-standard uses for ordinary objects, such as bubble wrap, frisbees and bricks. After consenting, participants were randomly assigned to one of three conditions that varied whether they completed the idea-generation task with or without access to an AI chatbot (please see Figure 1) for the stimuli used in each condition). The high-agency mindset and low-agency mindset conditions provided participants with access to a custom AI chatbot that was designed for the study, which was built on top of gpt-3.5 and embedded into the survey through a web-application. In the high-agency condition, participants were instructed to use AI to expand

Fig. 1. Experimental Paradigm



Note: Submitters were randomly assigned to one of three between-subjects conditions. In the high- and low-agency conditions, participants interacted with an AI chatbot to develop their creative ideas, as shown in an example trial on the right.

upon their own ideas and generate ones that are even more creative. This aspect of the manipulation was informed by prior research on growth mindsets [20].

Conversely, participants in the low-agency condition were told that "AI recognizes patterns, styles, and connections that would take years of study to master on your own... The beauty of AI is that it does the heavy lifting for you." They were instructed to let AI's creative power work for them. In the control condition, participants completed the same AUT without access to an AI chatbot. Given that past work suggests that online survey respondents frequently use AI tools to generate answers [18], we explicitly instructed participants in the control condition to not use AI in this task. Before the start of the AUT, participants read, "We value your individual human creativity, so please do NOT use AI during the task." Additionally, copy-paste functionality was disabled using JavaScript code embedded in the Qualtrics survey. Participants were required to spend a minimum of 10 seconds on the instruction page in each condition before proceeding to the AUT.

For each of three objects, frisbee, bubble wrap, and brick, submitters were asked to generate five alternative uses for the object and enter their ideas in form fields. The presentation of the three objects was counterbalanced. Participants in the high-agency and low-agency conditions had access to ChatGPT through an embedded web-app in the Qualtrics survey; they entered their alternative uses below the web-app interface. Participants in the control condition had no web-app, just the fields to enter their alternative uses for each object. In line with prior research on induction with the AUT [11], all participants were given as much time as they needed to complete the task. This avoided penalizing

the participants in the low-agency and high-agency conditions for taking time to work with the AI chatbot. Across conditions, submitters produced a total of $n = 2,250$ ideas.

Finally, all submitters provided a brief description of their idea generation process in a text box. Participants in the high-agency and low-agency conditions were asked to self-report the extent to which the AI helped them with the creative process (1 = Strongly Disagree to 7 = Strongly Agree; [10], the extent to which they perceived AI as a conversational partner (1 = Strongly Disagree to 7 = Strongly Agree; [10], and their intrinsic motivation during the AUT (how interesting, enjoyable, fun, and engaging it was; 1 = Not at all to 7 = Completely [17]). All participants answered demographic questions before being debriefed on the purpose of the study and granted research credits for their participation.

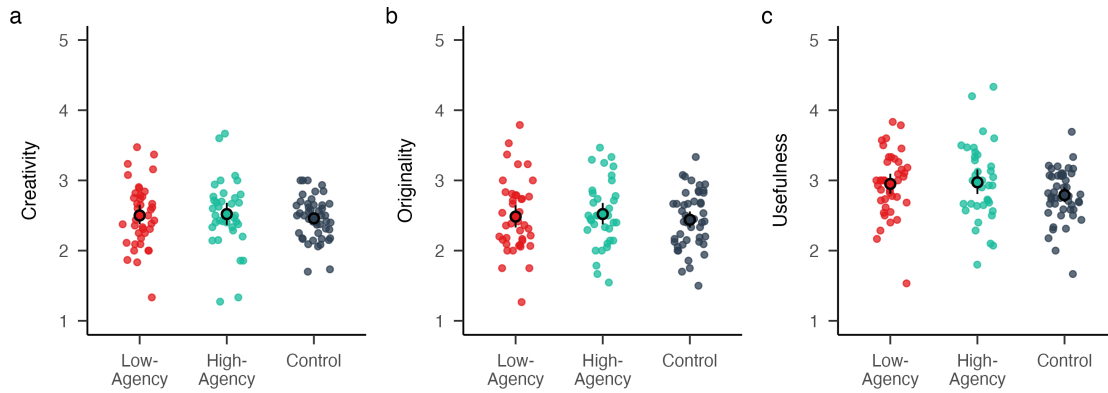
2.2.2 Evaluation Phase. Next, we asked a group of independent evaluators to rate the ideas submitted during the submission phase. These participants were shown the same definitions of creativity, originality, and usefulness as the submitters, which served as their rubric upon which to score the ideas. Each evaluator rated a random subset of 15 of the total set of compliant ideas ($n = 1785$). The submitted ideas were stratified by object type, such that each evaluator rated five ideas for bubble wrap, five ideas for a brick, and five ideas for a frisbee. For each of the 15 ideas they saw, the evaluators rated it in terms of overall creativity, originality, and usefulness, resulting in 45 ratings produced by each evaluator. Due to our randomization procedure, some ideas were evaluated multiple times, while others ($n = 76$) were never evaluated and thus were excluded from this analysis. All participants answered demographic questions before being debriefed on the purpose of the study and compensated for their participation.

2.2.3 Manipulation Checks. We conducted manipulation checks to ensure that participants followed the instructions for their conditions. Using the chat log data, we identified trials in which participants in the high-agency and low-agency treatment conditions did not use the AI tool as instructed. In 24% of low-agency trials and 38% of high-agency trials participants did not use the AI tool. Trials from the high- and low-agency conditions were excluded if they did not include chat data. On the participant level this resulted in 12 participants in the high-agency condition and 10 participants in the low-agency condition being completely excluded from the analysis. The remaining sample of submitters was composed of 38 participants in the high-agency condition, 40 participants in the low-agency condition, and 50 participants in the control condition.

2.2.4 Analytic Approach. We assessed the creative qualities of participants' submitted ideas at both the individual level and the condition level. To assess creativity, we examined the ratings from evaluators at the individual participant level. To assess homogeneity, we first used the sentence embedding model all-MiniLM-L6-v2 [16] to generate embeddings in a 384-dimension embedding space. Using these embeddings, we used hierarchical clustering to identify categories of ideas, the proportion of ideas that did not fall cleanly into any category, and the relative sizes of clusters.

2.2.5 Data Structure. Each participant produced five ideas for three objects (brick, bubble wrap, frisbee). For participants in the high-agency and low-agency conditions, the survey was designed such that each object corresponded to one human-AI interaction, with chat data logged using JavaScript.

Fig. 2. Individual-Level Outcomes Across Conditions



Note: Evaluators' ratings are displayed as jittered points, with condition means overlaid as larger points. Error bars are bootstrapped 95% confidence intervals.

3 Results

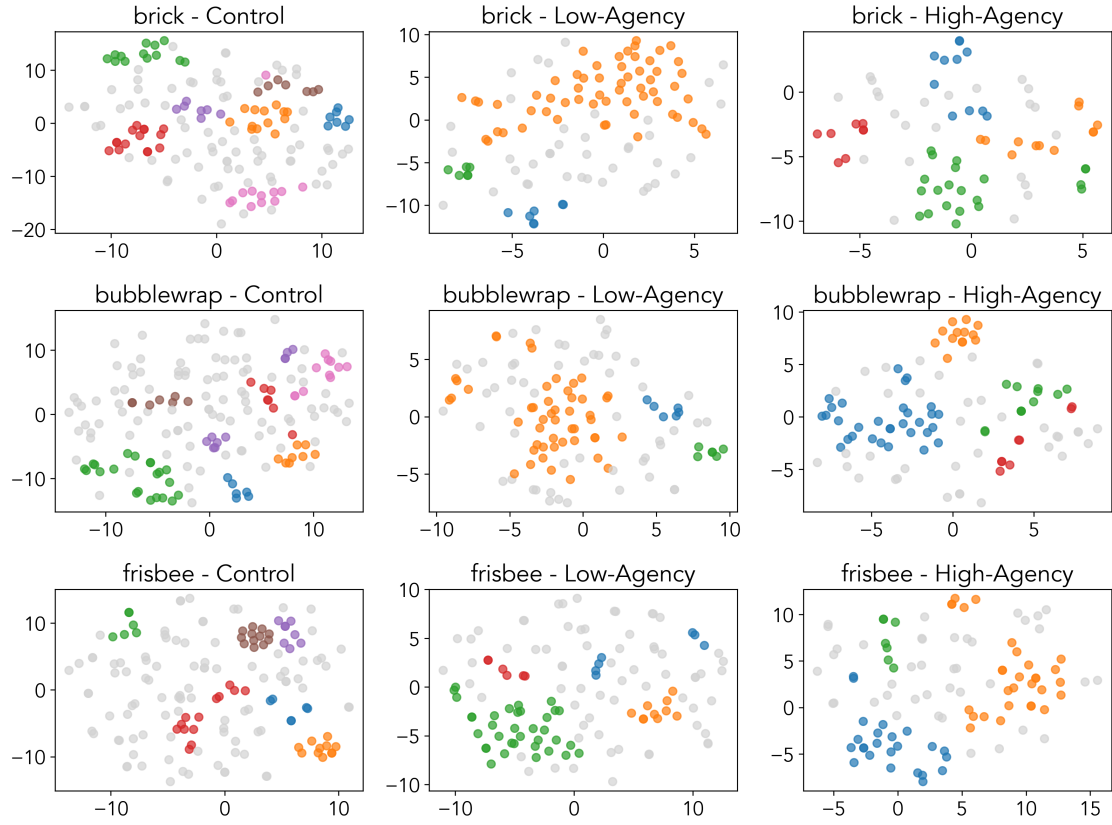
3.1 Individual-Level Outcomes

To examine the creativity of the submissions, we conducted separate mixed-effects linear regression models to predict each of the main dependent variables (creativity, originality and usefulness ratings from third-party evaluators) from the fixed effect of condition, with random effects for submitters and objects (bubble wrap, brick, frisbee). Contrary to our expectations, there were no significant differences in overall idea creativity or originality across the mindset conditions relative to control, $p_s > .60$, as shown in Figure 2.

3.2 Homogeneity

As an exploratory analysis of group-level homogeneity, we used hierarchical clustering via HDBSCAN [13] to identify categories of ideas proposed for each combination of object. We used three metrics to quantify the results of the clustering analysis: (1) number of unique clusters, (2) noise ratio, and (3) relative cluster size. The number of unique clusters serves as a proxy for the number of unique categories of uses proposed by participants in each condition. Noise ratio is the proportion of ideas that did not fall under any cluster. The coefficient of variation (CV; standard deviation of cluster sizes divided by the mean cluster size) indicates the extent to which the clusters are equal in size; a higher CV suggests greater imbalance in cluster sizes. One-way ANOVAs revealed significant effects of condition on all three clustering metrics. As shown in Figure 3, the control condition ($M = 6.67$, $SD = 0.58$) produced the greatest number of unique clusters, significantly exceeding the high-agency ($M = 3.67$, $SD = 0.58$) and low-agency conditions ($M = 3.33$, $SD = 0.58$), $F(2, 6) = 30.33$, $p < .001$. The control condition also had the greatest noise ratio ($M = 0.57$, $SD = 0.06$), followed by the low-agency condition ($M = 0.45$, $SD = 0.09$) and the high-agency condition $M = 0.35$, $SD = 0.05$), $F(2, 6) = 7.36$, $p < .03$. Notably, in the low-agency condition a single cluster tended to dominate the set of ideas for a given object, as indicated by the coefficient of variation ($M = 0.95$, $SD = 0.11$) significantly exceeding that of the high-agency ($M = 0.47$, $SD = 0.09$) and control condition ($M = 0.36$, $SD = 0.10$), $F(2, 6) = 27$, $p > .001$.

Fig. 3. Clusters Across Conditions and Objects



Note: Clusters are visualized in 2D space using t-distributed stochastic neighbor embedding (t-SNE). Each unique cluster is shown in its own color for each condition-object pair. Ideas that were classified as "noise" by the clustering algorithm appear in light gray.

4 Discussion

In the present study, we investigated the relationship between mindsets about AI usage and creativity at both the individual and group levels. Although we observed no differences in individual-level creativity via subjective assessments, our computational group-level analysis revealed distinct patterns of clustering. Within our sample, the condition generating ideas without AI assistance produced a higher number of unique categories of ideas. Furthermore, participants in the low-agency condition tended toglom onto a single cluster, while this effect was attenuated among the high-agency condition.

4.1 Limitations

Notably, a high rate of noncompliance was observed in the AI-access conditions. In their feedback, some participants mentioned refusing to use AI for creative tasks on principle. Such psychological reactance would be in line with prior research on the forced adoption of self-service technology [4] or the broader phenomenon of aversion to AI in

conventionally human domains [2]. While this somewhat compromised the random assignment, it also revealed real reactions to AI and digital literacy guidance that may occur in the wild, especially for a context as central to human identity as creation [14].

4.2 Implications

AI systems have lowered the barrier to entry for many forms of creative work and expression. Our study suggests that without an agentic mindset people who use large language models may unwittingly contribute to an algorithmic monoculture. Aside from broad threats to the creative thinkers whose work comprises the model’s training data, mass reliance on the same model may have a more insidious effect on the everyday users who count on an idea being just a prompt away. We posit mindsets as one route to preserving the diversity of human ideas and call upon future research to test additional interventions.

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